

Muawiya Amir

Software Engineer · Backend, ML & Open Source

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PROFESSIONAL SUMMARY

Artificial Intelligence and Mathematics student and active open-source contributor. Triage Collaborator on `drt-hub/drt`, a Python Reverse ETL engine shipped across 30+ PyPI releases, where I build connectors, configuration validation, and observability tooling and review community pull requests. I work across backend systems, machine-learning infrastructure, and production open-source tooling, and I am drawn to problems that need both engineering depth and clean execution.

TECHNICAL SKILLS

Languages: Python, C / C++, JavaScript, TypeScript, Java, SQL, x86_64 Assembly

AI / ML: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Computer Vision, NLP

Web & Backend: React, Node.js, Flask, REST APIs, HTML / CSS

Systems & DevOps: Git, GitHub Actions, Docker, CI/CD, pytest, JSON Schema, OpenTelemetry

CS Foundations: Operating Systems, Data Structures, Algorithms, Linear Algebra, Probability

PROFESSIONAL EXPERIENCE

Open-Source Engineer & Triage Collaborator, `drt-hub/drt`

Apr 2026 — Present

- Promoted to Triage Collaborator after sustained, high-quality contributions; review pull requests, triage issues, and shape feature design with maintainers.
- Designed and built the dynamic connector registry that auto-discovers sources and destinations, removing hardcoded dispatch maps across the engine.
- Added the Jira connector (create/update), JSON Schema validation for config files, and data-quality tests (freshness, unique, `accepted_values`).
- Shipped OpenTelemetry tracing and metrics with lazy, idempotent providers (+680 lines) and dry-run row-count diffs for safe sync previews.

Full-Stack Developer (Part-Time), SudoStudy

Sep 2025 — Present

- Build backend services and databases for an education platform and integrate third-party APIs.
- Shipped student features including native mobile-camera capture for long-answer questions and quiz-attempt review for teachers.
- Debugged cross-stack failures across quiz, subject, and group-management flows.

PROJECTS

Linear Algebra Library — A Python numerical library built only from primitive types — no math libraries — implementing core linear-algebra routines by hand.

SudoStudy — Backend services and mobile-facing features for a live education platform, including native camera capture for long-answer questions.

BSAI University Projects (NFC-Projects) — Structured collection of semester-wise coursework and systems projects for my BS in Artificial Intelligence at NFC IET Multan. Includes data structures, algorithms, IoT simulators, search engine implementations, and AI academic tools.

Diskern — Intelligent, read-only disk cleaner and analyzer that scans local file systems to determine file safety before freeing space. Built with a deterministic rules engine in Rust (`crates/diskern-core`) and a Tauri v2 + React desktop interface.

OPEN-SOURCE CONTRIBUTIONS

- **`drt-hub/drt`** — Triage Collaborator on a Python Reverse ETL engine. Designed the dynamic connector registry, added connectors and config validation, and shipped observability support.
- **`apache/hugegraph-ai`** — Upgraded the Python client CI from 1.3.0 to 1.7.0 and migrated from manual Docker runs to GitHub service containers with health checks.
- **`linuxfoundation/crowd.dev`** — Audited and corrected outdated technical documentation, Node version, and repository URLs across the project.

EDUCATION

BS Artificial Intelligence

NFC Institute of Engineering & Technology, Multan

BS Mathematics

Virtual University of Pakistan

CERTIFICATIONS & COMPETITIVE PROGRAMMING

- Applied Machine Learning — Verified

- Python AI Developer — Mimo
- Google Cloud Specialization — NAVTTC (ongoing)
- 400+ problems solved on LeetCode, focused on DSA & algorithm optimization.